

02_eda.py

```
1  """
2  02_eda.py
3
4  Health Insurance Cost & Claims Analytics
5  Exploratory Data Analysis (EDA)
6
7
8  Run this after 01_data_cleaning.py. re already
9  available in the session/kernel.
10 """
11
12
13 # EDA – distributions
14
15
16 numeric_cols = ["age", "bmi", "bloodpressure", "children", "claim"]
17 df[numeric_cols].hist(bins=20, figsize=(12, 8), color="skyblue", edgecolor="blue")
18 plt.suptitle("Distribution of Numerical Features", fontsize=16)
19 plt.tight_layout()
20 plt.show()
21
22 cat_cols = ["gender", "diabetic", "smoker", "region"]
23 plt.figure(figsize=(12, 8))
24 for i, col in enumerate(cat_cols, 1):
25     plt.subplot(2, 2, i)
26     sns.countplot(data=df, x=col)
27     plt.title(f"Distribution of {col}")
28 plt.tight_layout()
29 plt.show()
30
31
32 # Calculates the average insurance claim amount grouped by gender and smoker status,
33 # rounded to 2 decimal places.
34
35 df.groupby(["gender", "smoker"])["claim"].mean().round(2)
36
37
38 plt.figure(figsize=(8,5))
39 sns.barplot(data=df, x="gender", y="claim", hue="smoker", estimator="mean", errorbar="sd")
40 plt.title("Average Insurance Claim by Gender and Smoking Status")
41 plt.show()
42
43
44 # EDA – claim by smoker / diabetic / region
45
46 print(df.groupby(["gender", "smoker"])["claim"].mean().round(2))
```

```
47
48 pivot_region_diabetic = df.groupby(["region", "diabetic"])["claim"].mean().unstack()
49 print(pivot_region_diabetic)
50
51 pivot_region_smoker = pd.pivot_table(df, values="claim", index="region",
52                                     columns="smoker", aggfunc="mean")
53 print(pivot_region_smoker)
54
55
56 # Visualizes the pivot table as a grouped bar chart, comparing average claim amounts
    across regions, split by diabetic status.
57
58
59 pivot_region_diabetic.plot(kind="bar", figsize=(8,5))
60 plt.title("Average Claim by Region & Diabetic Status")
61 plt.ylabel("Mean Claim")
62 plt.show()
63
64
65 # Interaction check called out in the report: diabetic status alone is
66 # a weak signal – look at it combined with smoker and BMI category
67
68
69 df["bmi_category_tmp"] = pd.cut(
70     df["bmi"], bins=[0, 18.5, 24.9, 29.9, 100],
71     labels=["underweight", "normal", "overweight", "obese"]
72 )
73 print(df.groupby(["diabetic", "bmi_category_tmp"])["claim"].mean().round(2))
74 print(df.groupby(["diabetic", "smoker"])["claim"].mean().round(2))
75 df = df.drop(columns=["bmi_category_tmp"])
76
77
78 # Correlation heatmap
79
80
81 plt.figure(figsize=(8, 6))
82 sns.heatmap(df[numeric_cols].corr(), annot=True, cmap="coolwarm", fmt=".2f")
83 plt.title("Correlation Heatmap")
84 plt.show()
85
86
87 # Visualizes the relationship between age and claim amount as a scatter plot, with color
    showing smoker status and marker shape showing gender.
88
89
90 sns.scatterplot(data=df, x="age", y="claim", hue="smoker", style="gender")
91 plt.title("Claim vs Age by Smoker & Gender")
92 plt.show()
93
```

```
94
95 # bmi and claim
96
97 #
98 sns.regplot(data=df, x="bmi", y="claim", scatter_kws={"alpha": 0.6})
99 plt.title("Relationship between BMI & Claim Amount")
100 plt.show()
101
102
103 # Visualizes the spread and distribution of claim amounts for each number of children
    (using a boxplot to show median, quartiles, and outliers).
104
105
106 sns.boxplot(data =df, x="children", y = "claim")
107 plt.title("claim distribution by children")
108 plt.show()
109
110
111 # Age group / BMI category for reporting
112
113
114 df["age_group"] = pd.cut(df["age"], bins=[0, 18, 30, 45, 60, 110],
115                          labels=["<18", "18-30", "31-45", "46-60", "60+"])
116 print(df["age_group"].value_counts())
117
118 df["bmi_category"] = pd.cut(df["bmi"], bins=[0, 18.5, 24.9, 29.9, 100],
119                             labels=["underweight", "normal", "overweight", "obese"])
120 print(df["bmi_category"].value_counts())
121
122
123 sns.boxplot(data=df, x="bmi_category", y="claim")
124 plt.title("Claim Distribution by BMI Category")
125 plt.show()
126
127
128 sns.barplot(data = df, x="age_group",y ="claim",estimator ="mean",errorbar ="sd")
129 plt.title("average claim by age group")
130 plt.show()
131
132
133
134
135
136 # Volume-vs-severity split flagged in the report: is the 31-45 / Southeast
137 # concentration a volume effect or a severity effect?
138
139
140 print(df.groupby("age_group")["claim"].agg(["count", "mean", "sum"]))
141 print(df.groupby("region")["claim"].agg(["count", "mean", "sum"]))
```

```
142
143
144 # Regional Summary: Smoker Rate & Average Claim
145 # Groups the data by region to calculate two key metrics: the percentage of smokers in
    each region, and the average claim amount per region.
146
147
148 region_stats = df.groupby("region").agg(
149     smoker_rate=("smoker", lambda x: (x == "yes").mean() * 100),
150     mean_claim=("claim", "mean")
151 ).reset_index()
152
153
154 region_stats
155
156
157 fig, ax1 = plt.subplots(figsize=(8,5))
158
159 sns.barplot(data=region_stats, x="region", y="smoker_rate", ax=ax1, alpha=0.6)
160
161 ax2 = ax1.twinx()
162 sns.lineplot(data=region_stats, x="region", y="mean_claim", ax=ax2, color="red",
    marker="o")
163
164 ax1.set_ylabel("Smoker Rate (%)")
165 ax2.set_ylabel("Average Claim ($)")
166 plt.title("Smoker Rate and Average Claim by Region")
167 plt.show()
168
```